

Perhaps the most-related work here is [Recchia \(2021\)](#), which shows that finetuning enables longhand module operations, which has previously been difficult for performers. Whereas work in this direction is often task-specific and uses finetuning, we show that chain-of-thought prompting works for a broad range of tasks without any finetuning.

C.5 Intermediate language steps

Extensive prior work has shown the benefits of endowing neural networks with the ability to produce intermediate steps via training or finetuning confers various benefits in a range of scenarios. As examples, it has been shown that natural language intermediate steps can improve performance ([Zaidan et al., 2007](#); [Yao et al., 2021](#); [Hase and Bansal, 2022](#); [Gu et al., 2022](#)), improve robustness ([Chen et al., 2022](#)), speed up training ([Hancock et al., 2018](#)), mitigate bias ([Dua et al., 2020](#)), and even help in image and reinforcement learning settings ([Andreas et al., 2018](#)). To endow models with the ability to produce intermediate steps, prior work typically finetunes models on either manually annotated training datasets ([Camburu et al., 2018](#); [Rajani et al., 2019](#), *inter alia*) or generates synthetic datasets ([Talmor et al., 2020](#); [Zelikman et al., 2022](#)). Compared with these training or finetuning methods, our work shows that various natural language reasoning abilities can be elicited in off-the-shelf language models of sufficient scale simply via prompting. This prompting setup is important because it allows for intermediate step reasoning without a large number of labeled annotations, and because a single model can perform a range of reasoning tasks without any gradient updates.

D Appendix: Additional Analysis

D.1 Correct Chain of Thought Analysis

As mentioned in the main text, we analyze 50 chains of thought from LaMDA 137B that led to correct answers in the GSM8K dataset. Of these 50, only one arrived at the correct answer through incorrect reasoning (shown in Table 9: “correct by chance”). The other 49 had correct logic and math, with examples shown in Table 8. Five had minor imperfections while maintaining coherent and understandable logic:

- One had underspecified statements (shown in Table 9: “correct but underspecified statements”).
- One made an unrelated but correct statement (shown in Table 9: “correct but unrelated statement”).
- Two had the correct logic but omitted specific steps in the equation, though it was still understandable (shown in Table 9: “correct but imperfect equation”).
- One had the correct math but inverted the semantics (shown in Table 9: “correct but inverted semantics”).

Table 8: Example of correct chains of thought produced by the model for the GSM8K dataset.

QUESTION: James decides to run 3 sprints 3 times a week. He runs 60 meters each sprint. How many total meters does he run a week?

MODEL ANSWER (CORRECT): James decides to run 3 sprints 3 times a week. He runs 60 meters each sprint. So he runs 60 meters x 3 sprints x 3 times a week. That is 60 meters x 9. The answer is 540. ✓

QUESTION: Brandon’s iPhone is four times as old as Ben’s iPhone. Ben’s iPhone is two times older than Suzy’s iPhone. If Suzy’s iPhone is 1 year old, how old is Brandon’s iPhone?

MODEL ANSWER (CORRECT): Brandon’s iPhone is 4 times as old as Ben’s iPhone. Ben’s iPhone is 2 times older than Suzy’s iPhone. So Brandon’s iPhone is $4 \times 2 = 8$ times older than Suzy’s iPhone. Suzy’s iPhone is 1 year old. So Brandon’s iPhone is $8 \times 1 = 8$ years old. The answer is 8. ✓

QUESTION: Jean has 30 lollipops. Jean eats 2 of the lollipops. With the remaining lollipops, Jean wants to package 2 lollipops in one bag. How many bags can Jean fill?

MODEL ANSWER (CORRECT): Jean started with 30 lollipops. She ate 2 of them. So she has 28 lollipops left. She wants to package 2 lollipops in one bag. So she can package $28 / 2 = 14$ bags. The answer is 14. ✓

Although we find that chain-of-thought reasoning is mostly correct for math questions where the final answer was correct, this is likely because it is hard to arrive at the correct final answer by coincidence for free response questions. For multiple choice or binary classification problems, it was much more